UW-Madison GI Tract Image Segmentation

Track Healthy Organs in Medical Scans to Improve Cancer Treatment

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Project Recap

The Challenge

This challenge hosted by the **UW-Madison Carbone Cancer Center**, aims to **automate the segmentation of the stomach and bowels** in MRI scans to enhance the precision of cancer radiotherapy.

- **Task:** Pixel-wise semantic segmentation of GI tract organs.
- **Output:** Binary masks converted to **Run-Length Encoding (RLE)** for efficiency.

The Importance

Enable **real-time adaptive radiotherapy**, allowing clinicians to deliver higher radiation doses to tumors while sparing healthy moving organs, ultimately reducing patient treatment time and toxic side effects.

Dataset Anatomy: Scale & Structure

Attribute	**Specification**
Total Images	38,496 MRI Slices
Case Studies	85 Patients (Patient Data Spans Multiple Days)
File Format	16-bit Grayscale PNG
Variable Resolution	e.g., 266×320 to 360×310 px
Total Size	2.47 GB

Table: Also Contains a csv of all IDs and Masks of training objects

DataLoader Outline

1. Initialization

```python
#initialization
def __init__(self, csvPath, folderPath, imgSize = 320, statsPath = None):
    self.df = pd.read_csv(csvPath) #train.csv
    self.folderPath = Path(folderPath) #train ~ the folder
    self.imgSize = imgSize #resolution size ~ can be changed, but defaulted to 320
```

- Loads `train.csv` via Pandas
- Maps the `/research` Directory

2. Path Navigation

```python
#fetches, processes, packages
def __getitem__(self, idx):
    row = self.df.iloc[idx] #grabs specific csv row

    #path navigation
    parts = row['id'].split('_') #splits string id
    caseID, dayID, sliceID = parts[0], parts[1], parts[3]
    folderPath = self.folderPath / caseID / f"{caseID}_{dayID}" / "scans"
    imgPath = list(folderPath.glob(f"slice_{sliceID}_*.png"))[0]
```

- Parses ID: Case/Day/Slice
- Dynamic `glob` path assembly

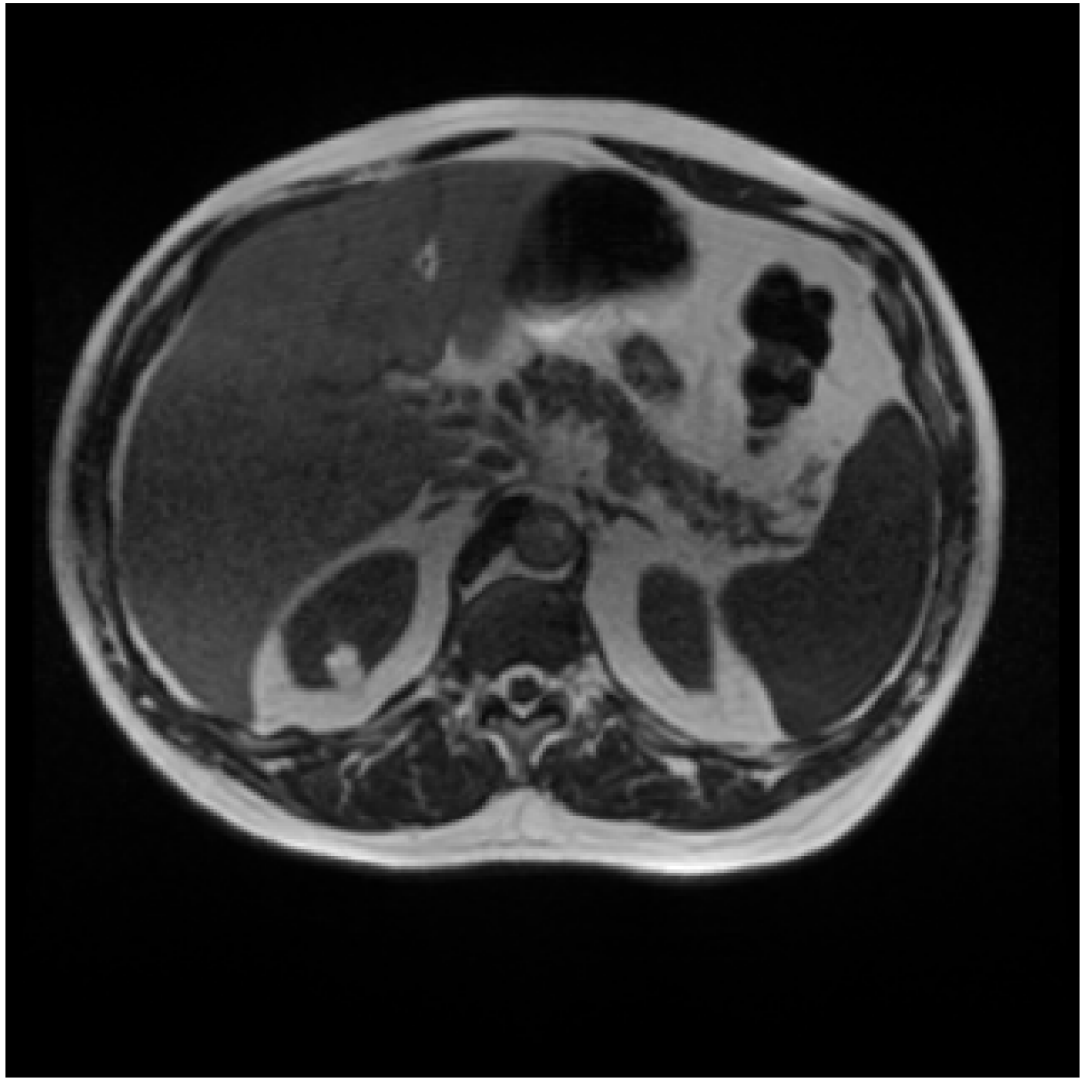
3. PyTorch DataLoader

```python
#training pass
trainDataset = GITractDataset(csvPath, folderPath, imgSize = 320, statsPath = "calcStats.json")
trainLoader = DataLoader(
    trainDataset,
    batch_size = 16,
    shuffle = True,
    num_workers = 4,
    pin_memory = True
)
```

- Batch size of 16 across 4 Workers
- Shuffle & Pin Memory enabled

DataLoader Verification

Raw MRI Scan (Slice 248)



Preprocessing Pipeline

1. Global Dataset Calibration

```python
#calibration
def calcStats(loader, savePath = "calcStats.json"):
    psum, psum_sq = 0, 0 #store sums
    count = 0 #store count

    #gather variables through data
    for imgs, _ in loader:
        psum += imgs.sum()
        psum_sq += (imgs**2).sum()
        count += imgs.numel()

    #maths
    mean = psum / count
    var  = (psum_sq / count) - (mean**2)
    std  = torch.sqrt(var)

    #stores data
    stats = {"mean": mean.item(), "std": std.item()}
    with open(savePath, "w") as f:
        json.dump(stats, f)

    return stats
```

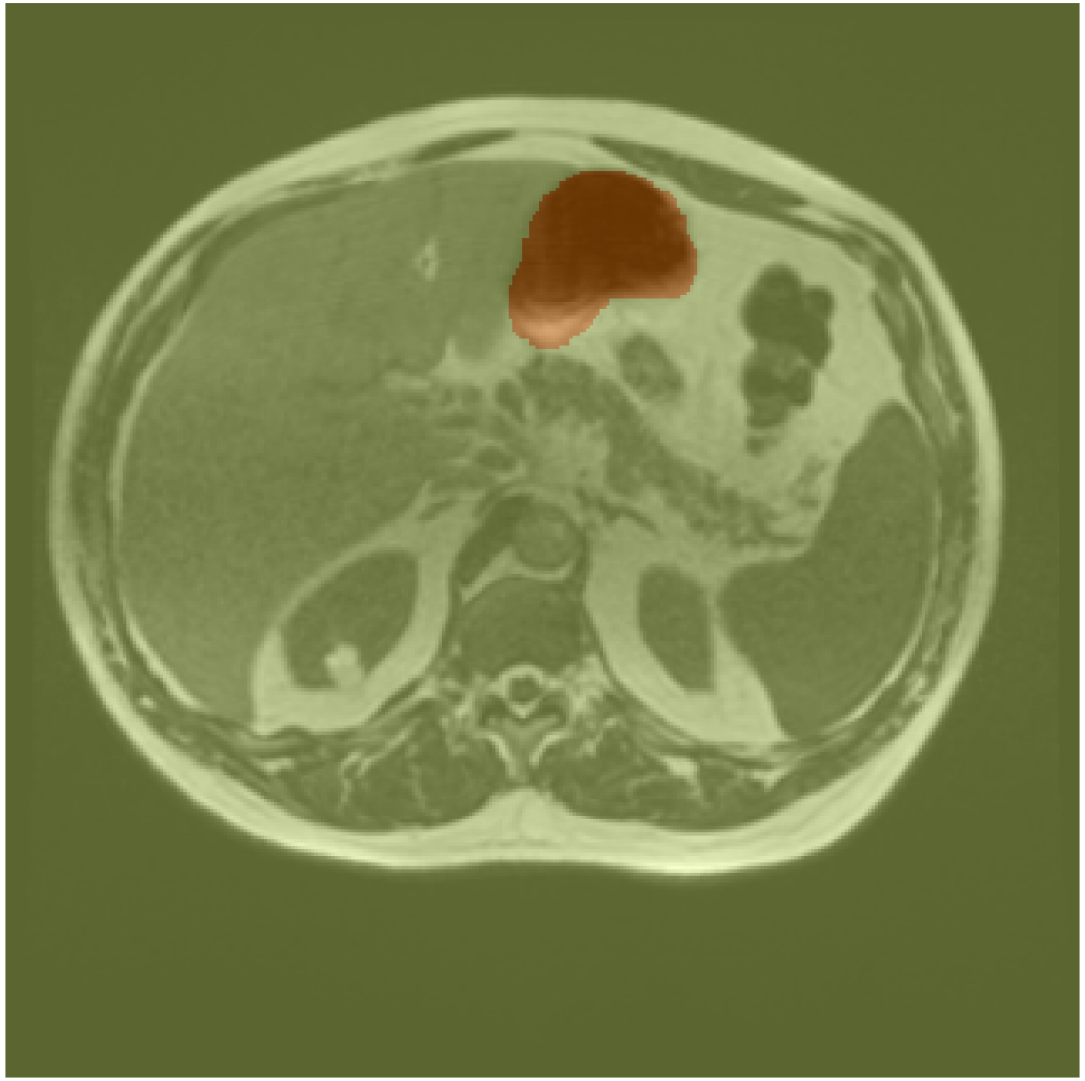
2. Scaling & Normalization

```python
#loading/preprocessing
img = cv2.imread(str(imgPath), cv2.IMREAD_UNCHANGED).astype(np.float32) #convert 16-bit to decimal
mask = self.rleDecoder(row['segmentation'], img.shape) #getting mask

#resizing
img = cv2.resize(img, (self.imgSize, self.imgSize))
mask = cv2.resize(mask, (self.imgSize, self.imgSize), interpolation=cv2.INTER_NEAREST)

#scale img between 0.0 to 1.0 ~ ensures model gradients remain stable
if img.max() > 0:
    img = img / img.max()

#normalization
img = (img - self.mean) / self.std
```

- Standardizes input to mean $= 0$ and standard deviation $= 1$ across the dataset
- Stores stats in `calcStats.json` for consistent normalization and de-normalization
- Resizing images to a uniform resolution

Preprocessing Verification

Normalized Scan with Segmentation Mask



Augmentation Plans

1. Intensity Histogram Scaling

- **Technique:** Contrast Limited Adaptive Histogram Equalization (CLAHE).
- **Goal:** Normalize varying signal intensities across different MRI scanners.

2. ROI Patch-Based Training

- **Technique:** Random cropping focused on regions with non-zero masks.
- **Goal:** AI to learn the texture of the intestines rather than just memorizing where they are in the frame.

Top Performance

Random Cropping: Scaled image sizes to 640*640 and then used random crop to target a 448*448 resolution. **Elastic Transform/Grid Distortion/Optical Distortion:** Used these series of methods to cause warping to identify organs even if stretched.

Evaluation Metrics: Overlap and Boundary Precision

1. Dice Coefficient (F1-Score)

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

- **Focus:** Measures the spatial overlap between the predicted mask (X) and ground truth (Y).
- **Justification:** Ideal for imbalanced data; it ignores true negatives (background) to focus purely on the GI tract pixels.

2. 3D Hausdorff Distance (HD)

$$d_H(X, Y) = \max\{\vec{d}_H(X, Y), \vec{d}_H(Y, X)\}$$

- **Focus:** Measures the maximum distance between the surface of the prediction and the ground truth.
- **Justification:** Crucial for clinical safety; it penalizes "outlier" pixels that are far from the actual organ boundary.

Future Work: Modeling Optimization

1. Architectural Evolution

- **U-Net Baseline:** Implement a standard 2D U-Net for initial segmentation.
- **Transition to 2.5D:** Stack adjacent slices (e.g., 3-5) as input channels to provide the model with "pseudo-3D" spatial context.

2. Data Enhancements

- **Augmentation:** Integrate random ROI cropping and CLAHE.

3. Performance Evaluation

- **Validation:** Utilize `calcStats.json` to standardize validation set to simulate real-world model deployment.
- **Metric Tracking:** Compare Dice and Hausdorff scores across different model iterations.

Note: 2.5D captures MRI depth context without the massive overhead of 3D convolutions.

Questions & Discussion